Problem 1 and Problem 2 a)

Exercise 7

1) a) S_{11} is the reflection coefficient for the two-port system input. How much the signal reflects back from port 1.

b) S_{21} is the forward transducer power gain. Measures the signal passing from port 1 to port 2

c) The S parameters are dependent on Z_0, like S_{11} that is $\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$. They are all dependent on Z_0 as we measure through the transmission to get the S-parameters.

d) In a lossless system, $|S_{11}|^2 + |S_{21}|^2 = 1$, meaning the magnitude of $|S_{11}|$ is decided by $|S_{11}| = \sqrt{1 - |S_{21}|^2}$ the power gain.

2) a) $Z_{in}(l) = Z_0 \frac{Z_L + jZ_0 \tan\left(\frac{2\pi}{\lambda} l\gamma\right)}{Z_0 + jZ_L \tan\left(\frac{2\pi}{\lambda} l\gamma\right)}$ | $\beta = \frac{2\pi}{\lambda}$, $l = l\gamma$

$$\Rightarrow Z_{in} = \frac{Z_L/50 + j\tan(2\pi\gamma)}{1 + Z_L/50\, j\tan(2\pi\gamma)}$$ | $= 6{,}48 + j0{,}44$

$Z_L/50 \Rightarrow Z_L \tan(2\pi\gamma) = TAN$

$\therefore (1 + Z_L/50\, j\tan(2\pi\gamma))^* / -1 -$

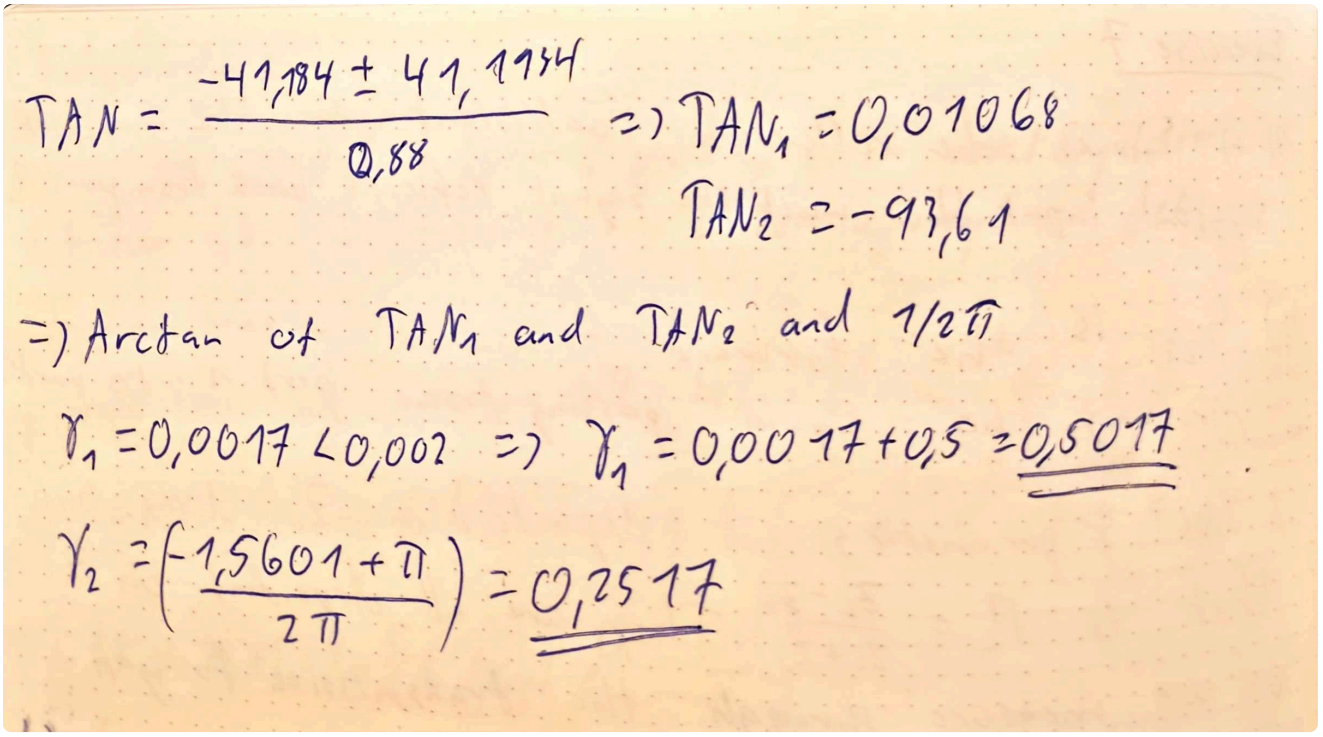
b) $Z_{in} = \frac{(Z_L + jTAN)(1 - jZ_L^* TAN)}{(1 + jZ_L TAN)(1 - jZ_L^* TAN)}$ | Im part needs to equal 0

$$\mathrm{Im}\{(Z_L + jTAN)(1 - jZ_L^* TAN)\} = 0$$

$$\mathrm{Im}\{Z_L - jZ_L^* Z_L TAN + jTAN + Z_L^* TAN^2\} = 0$$

$$\Rightarrow 0{,}44 - (6{,}48^2 + 0{,}44^2)TAN + TAN - 0{,}44TAN^2 = 0$$

$$\Rightarrow 0{,}44TAN^2 + 41{,}84TAN - 0{,}44 = 0$$ | solve for TAN



Problem 2)

b)

L:=324+ i*22

→ $L := 324 + 22\,i$

Z(a):=50 * ((L+ i*50*tan(2*pi*a))/(50 + i*L*tan(2*pi*a)))

→ $Z(a) := 50 \cdot \frac{50\,i\,\tan(2\,a\,\pi) + 324 + 22\,i}{(-22 + 324\,i)\,\tan(2\,a\,\pi) + 50}$

Z(0.2517)

≈ $7.68 - 0\,i$

Z(0.5017)

≈ $325.53 + 0\,i$

c)

50**2/7.68

≈ 325.52

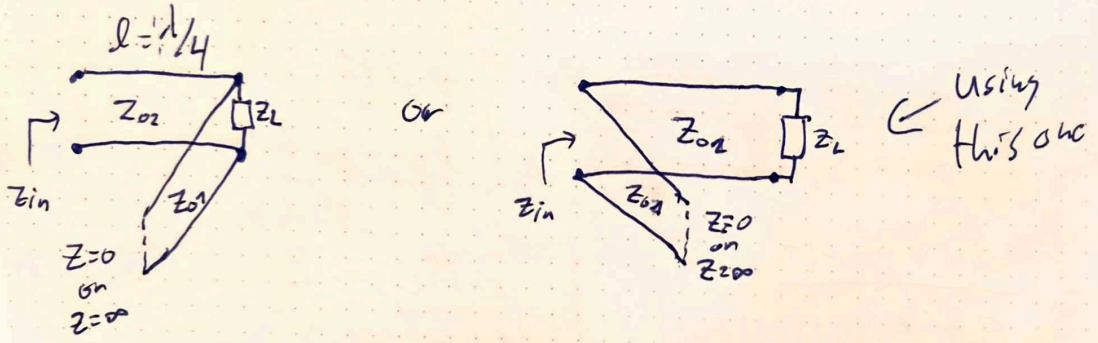
50**2/325.53

≈ 7.68

Seems pretty good match to me.

d & e)

d) We can place the stub either by the load or by the entrance like so:



1: Z_{01} designed so $jB_L = -jB_{01}$
for $Y = \frac{1}{Z} = G + jB$

2: λ/4 line transformer
so Z_{in} becomes real

e) Parallell are better for microstrip as it shares the main lines grounding plane and is thus easy to fabricate. You also dont have to break or interrupt the main line. And can easily make short or open stubs by adding or not a via to ground.

f)

f) $Z_L = 324 + j22\ \Omega \quad Z_0 = 50\Omega \quad \Rightarrow \frac{Z_L}{Z_0} = 6{,}48 + j0{,}44$

$Y_L = \frac{1}{Z_L} = \frac{R_L - jX_L}{R_L^2 + X_L^2} \Rightarrow y_L = \frac{Y_L}{Y_0} = g_L + jB_L$

$$y(l) = 1 \cdot \left[\frac{y_L + j\tan(\beta l)}{1 + j y_L \tan(\beta l)}\right] \Rightarrow Re\{y(l)\} = 1$$

$$\Rightarrow Re\left\{\frac{(y_L + j\tan(\beta l))(1 - j y_L^* \tan(\beta l))}{1 + (g_L^2 + B_L^2)\tan(\beta l)^2 - 2B_L\tan(\beta l)}\right\} = 1$$

$$\Rightarrow Re\left\{\frac{(g_L + jB_L + j\tan(\beta l))(1 - jg_L\tan(\beta l) - B_L\tan(\beta l))}{\quad}\right\} = 1$$

$$\Rightarrow \frac{g_L - g_L B_L \tan(\beta l) + g_L B_L \tan(\beta l) + \tan^2(\beta l) g_L}{1 - 2B_L\tan(\beta l) + g^2\tan^2(\beta l) + B_L^2\tan^2(\beta l)} = 1 \;\Big|\; t = \tan(\beta l)$$

$$\Rightarrow g_L(1 + t^2) = 1 - 2B_L t + (g_L^2 + B_L^2)t^2$$

$$\Rightarrow (g_L^2 + B_L^2 - g_L)t^2 - 2B_L t + (1 - g_L) = 0$$

$$\Rightarrow t = \frac{B_L \pm \sqrt{g_L((g_L - 1)^2 + B_L^2)}}{g_L^2 + B_L^2 - g_L} \;\Big|\; Y_L = 50\left(\frac{324 - j22}{324^2 + 22^2}\right) = 0{,}1536 - j0{,}0104$$

$\Rightarrow t_1 = -2{,}4402$ & $t_2 = 2{,}6002$

$\Rightarrow d_1 = 0{,}3111\lambda$ & $d_2 = 0{,}1923\lambda$ Distance to stub

$Y(d_1) \approx 1 - \underset{b_1}{2{,}1597} \quad Y(d_2) \approx 1 + \underset{b_2}{2{,}1597}$

$Y_{stub} = Y_{in,open} = j\tan(\beta l) = jb_{stub}$

$b_{stub} = -b_1 \Rightarrow l_1 = 0{,}1810\lambda$ length stub

$b_{stub} = -b_2 \Rightarrow l_2 = 0{,}3190\lambda$ stub